Check for updates

ASME Journal of Engineering for Gas Turbines and Power
Online journal at:
https://asmedigitalcollection.asme.org/gasturbinespower

Hari Priya Rajagopalan[1]
Ben T. Zinn Combustion Lab,
Georgia Institute of Technology,
Atlanta, GA 30318
e-mail: hrajagopalan3@gatech.edu

Ben Emerson
Ben T. Zinn Combustion Lab,
Georgia Institute of Technology,
Atlanta, GA 30318
e-mail: bemerson@gatech.edu

Vishal Acharya
Ben T. Zinn Combustion Lab,
Georgia Institute of Technology,
Atlanta, GA 30318
e-mail: vishal@aerospace.gatech.edu

David Noble
EPRI,
Charlotte, NC 28262
e-mail: bnoble@epri.com

Tim Lieuwen
Ben T. Zinn Combustion Lab,
Georgia Institute of Technology,
Atlanta, GA 30318
e-mail: tim.lieuwen@aerospace.gatech.edu

Measurements of CO and NO_x Emissions From Premixed Turbulent Methane/Hydrogen Flames

This paper presents measurements of the sensitivity of NO and CO emissions from H_2/CH_4 combustion to variations in turbulence intensity. Existing studies of exhaust emissions either focus on laminar modeling and measurements or report device-level measurements from practical combustors. However, there are limited fundamental studies of emissions characteristics of turbulent systems, the focus of this study. Because of the strong stretch sensitivity of these mixtures, local temperatures/reaction rates along the turbulent front can differ markedly from mixture-averaged properties, and they are very sensitive to turbulence. Single-point emissions measurements are presented for turbulent premixed H_2/CH_4 flames from a contoured nozzle Bunsen burner with variable turbulence. Emissions are measured at varying residence times and incoming flow turbulence intensities. Data show the clear benefit of H_2 blending on CO emissions, both due to the expected reduction in C atoms, but also due to faster relaxation of flame generated CO toward equilibrium. It also shows negligible impact of turbulence intensity on NO emissions at constant CO for CH_4.
[DOI: 10.1115/1.4069625]

Keywords: NO, CO emissions, H_2/CH_4 blends, gas turbine relevant conditions, premixed turbulent flames

1 Introduction

Development of hydrogen combustion strategies for gas turbine combustors is critical to achieving net zero emission goals set for 2050 [1]. Hydrogen-blended methane combustion systems present unique design challenges, including achieving a broader operability range with higher turndown, minimizing emissions, and implementing effective flashback prevention mechanisms [2,3]. These challenges arise from the significant differences in fundamental properties between hydrogen and methane fueled systems, such as flame propagation speed, stretch sensitivity, and adiabatic flame temperature [4–6] at the same equivalence ratio. Lean H_2 flames exhibit high stretch sensitivity, with local flame temperatures exceeding mixture-averaged values by over 200 K at points of strong positive curvature [7]. For this reason, local laminar burning velocities can surpass unstretched values by an order of magnitude or more [8]. NO and CO emissions are byproducts of hydrocarbon combustion. Hydrogen-blended premixed flames offer a notable advantage in reducing CO emissions. For instance, Singh et al. [9] reported a significant decrease in CO levels as the H_2 concentration in H_2/CH_4 blends increased. This reduction was attributed not only to the lower atomic carbon content in the fuel but also through increased relaxation rates of superequilibrium CO levels toward its equilibrium. It is likely that these relaxation rates would be further influenced by turbulence levels, as H_2-blended flames show strong sensitivity to turbulence [10–12], but we are not aware of studies of this nature. Two primary contributors to NO_x production in hydrocarbon flames are (1) flame or prompt NO_x—formed within the exothermic region of the flame through various fuel-based reaction pathways, and (2) thermal NO_x—produced in the postflame region with sustained high temperatures. Numerous studies have been performed to understand the effect of H_2 levels on the NO_x emissions [13–15]. First, it is important to emphasize that using volumetric measures (like ppmvd) are problematic for comparing across variable H_2 composition, and comparisons are better done using metrics like mass NO_x produced per unit of heat input [16]. In addition, it is crucial to carefully consider what else is being changed with H_2 content, particularly if adiabatic flame temperature is being changed, since H_2 has a higher adiabatic flame temperature as compared to methane for the same equivalence ratio. For example, some studies report a rise in NO_x levels as the H_2 content in fuel increases [17,18], which probably reflects this thermal influence. What is of more interest for premixed applications is H_2 effects at constant temperature. A detailed numerical analysis by Breer et al. [19] at constant adiabatic flame temperature explored the influence of hydrogen addition on NO_x emissions in a 1D flat flame and found that NO emissions at a fixed temperature were relatively insensitive to H_2 content, although it could go slightly up or down depending upon pressure and temperature. As practical combustion systems

Turbo Expo: Turbomachinery Technical Conference & Exposition (GT2025), June 16–20, 2025. GT2025.
[1]Corresponding author.
Manuscript received July 31, 2025; final manuscript received August 6, 2025; published online October 31, 2025. Editor: Jerzy T. Sawicki.

Copyright © 2026 by ASME; reuse license CC-BY 4.0

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/4/041004/7534825/gtp-25-1459.pdf by guest on 31 December 2025

operate in highly turbulent conditions, understanding the impact of incoming flow turbulence on flame structure and chemical reactions, which in turn influence flame emissions, is crucial. Numerical study [20] performed using perfectly stirred reactors with syngas fuel and coupled with a turbulent mixing submodel found that increased mixing frequencies amplify both CO and NO levels. This was primarily associated with the influence of mixing frequencies on oxyhydrogen radicals. Experimental investigations using low-swirl burners fueled by syngas and operating at gas turbine relevant conditions [21] showed a slight increase in NO_x levels as hydrogen content in the fuel increased. Furthermore, numerical studies comparing NO_x production between laminar and turbulent counterflow configurations fueled by syngas found that turbulent counterflow systems produced significantly lower NO_x levels than the laminar case for the same fuel consumption [22]. This reduction in NO_x levels was attributed to turbulence-induced enhancements in heat and mass transfer, which are hypothesized to suppress NO_x production rates. Direct numerical simulations by Day et al. [23] noted the influence of the cellular structure of lean H_2 flames on NO_x production, emphasizing the role of localized hot spots formed within these flames. These hot spots, exhibiting temperatures well above the average adiabatic flame temperature, arise due to enhanced preferential diffusion in lean premixed H_2 flames. The study identifies these hot spots as a contributor to elevated NO production, with the effect being particularly pronounced at leaner equivalence ratios where NO production increased by nearly threefold compared to laminar flame counterparts. At higher equivalence ratios, the relative variation from laminar flames was much smaller, likely due to the rapid mixing of these elevated-temperature regions, driving them closer to the mixture-averaged temperature. Given the varied results observed regarding the effects of H_2 blending and flow turbulence on NO_x and CO emissions, this paper aims to address these influences individually. To facilitate meaningful comparisons across H_2 blends, the data are constrained by maintaining a constant adiabatic flame temperature by varying $H_2\%$ and equivalence ratios simultaneously. This study has two primary objectives: first, to assess the impact of incoming flow turbulence on NO_x and CO emissions from turbulent premixed flames, and second, to analyze the influence of H_2 content in fuel on emissions from these flames.

2 Materials and Methods

Measurements of CO and NO_x emissions from atmospheric premixed turbulent hydrogen blended methane flames were made in a piloted Bunsen burner configuration with a nozzle diameter (D) of 12 mm and variable turbulence generator (see Fig. 1). This system comprises of H_2/CH_4 fuel metered with critical orifices and premixed with air two meters upstream from the point of ignition, enabling sufficient premixing of reactants. A premixed H_2/air annular pilot was used to stabilize the flame. The turbulence generator system comprises two radially slotted plates, enabling the adjustment of turbulence intensity by altering the blockage between them. The blockage ratio ranges from 69% to 97%, corresponding to angular slot openings from 30 deg to 2 deg. This system allows for independent control of bulk flow velocity and turbulence intensity while being remotely operable, making it a critical design element of the facility. Flow characteristics under isothermal conditions were assessed using laser Doppler velocimetry, with detailed findings reported in a previous study [24]. A quartz tube, 305 mm in length with an inner diameter of 52 mm and a wall thickness of 3 mm, was mounted at the nozzle exit to prevent dilution of the product gases with ambient air. Product gas samples were extracted using a clear fused quartz probe with the following dimensions: 4 mm ID, 6.3 mm outer diameter, and 100 mm length, tapered to a 1.5 mm diameter opening. This tapered design facilitates aerodynamic quenching to suppress chemical reactions in the sampled product gases [25]. The probe was mounted on a traverse system, allowing collection of gas samples at various axial locations, corresponding to different residence times. The acquired samples were then transported to

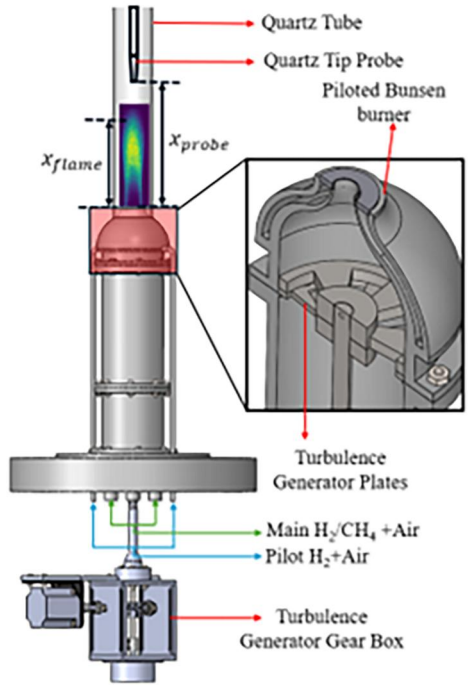

Fig. 1 Turbulent Bunsen burner setup

Testo 350 emissions analyzer via a gas filter and dryer. The dryer was used in line to dry the product gas sample before passing it to the analyzer, as the presence of water in the sample is detrimental to both the functioning of the analyzer and the accuracy of the measurements. The Testo 350 system is a portable fast exhaust gas analyzer that offers high-accuracy measurements of NO_x and CO emissions, with an uncertainty of ± 2 ppm below a measured value of 40 ppm and $\pm 5\%$ of measured value above it. It has a full-scale range of 300 ppm for NO_x and 500 ppm for CO. Table 1 lists the range of test conditions from this study.

2.1 Equilibrium Properties. Equilibrium flame properties, such as Tad and equilibrium product gas composition, were computed using CANTERA's chemical equilibrium solver [26]. The results from these simulations were postprocessed using PYTHON for further analysis.

2.2 Experimental Residence Time Calculation. To compare fundamental factors influencing emissions levels across test conditions, it is important to account for variations in flame height, which change postflame gas residence time if the probe location is fixed. Flame height is a function of turbulence intensity as well as reactant composition. The average flame height, defined as x_{flame}, is extracted using OH* chemiluminescence images, obtained with NatGX3 camera with videoscope intensifier at 200 Hz. A 50 mm f/2.8 Circo UV lens with OH filter of 308 ± 20 nm is used for these measurements. The acquired images are temporally averaged and processed using a 3×3 Gaussian filter. The resulting averaged image is then symmetrized by left/right averaging to produce an

Table 1 Test conditions

Parameters	Values
$H_2\%$	0–50
φ	0.65–0.897
T_{ad} (K)	1800–2150
U_0 (m/s)	10
u_{rms} (m/s)	1.15–2.5
P (bar)	1
T_{in} (K)	300

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/4/041004/7534825/gtp-25-1459.pdf by guest on 31 December 2025

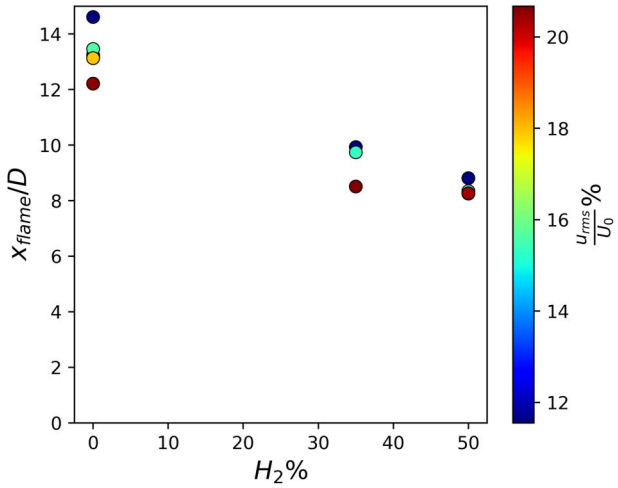

Fig. 2 Effect of H_2 and turbulence intensity on flame height (normalized by nozzle diameter) at fixed $T_{ad}=1800\pm20$ K

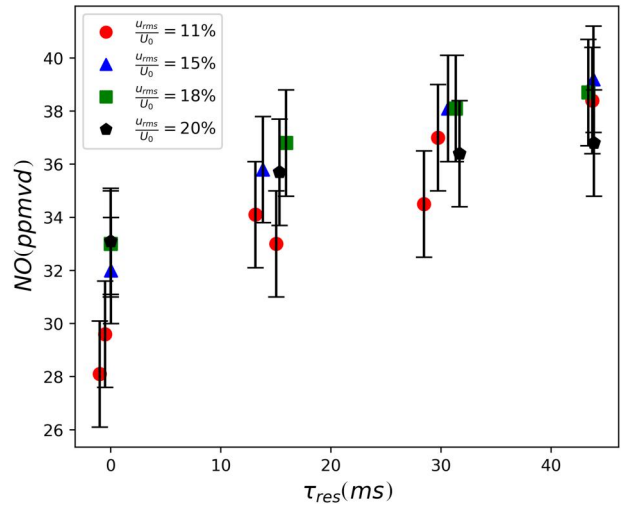

Fig. 3 NO levels as a function of residence time for CH_4–air flame at $\varphi=0.9$ and $U_0=10$ m/s across turbulence intensity

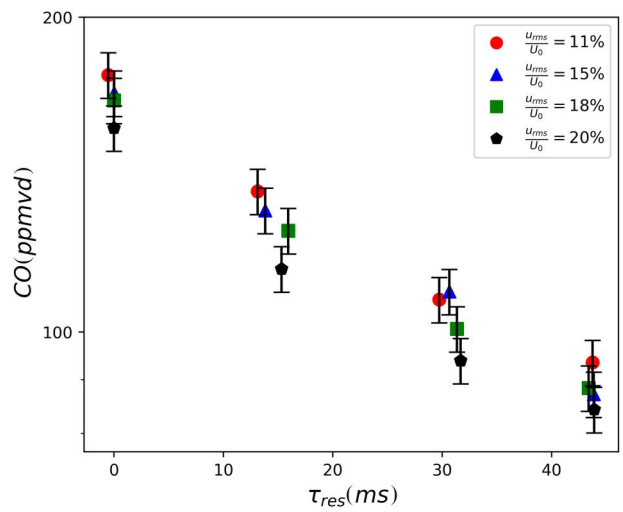

Fig. 4 CO levels as a function residence time for CH_4–air flame at $\varphi=0.9$ and $U_0=10$ m/s across turbulence intensity

axisymmetric flame representation. To extract central-plane flame images from the line-of-sight integrated chemiluminescence data, an inverse Abel transform is applied to the axisymmetric image. The flame height (x_{flame}) is then determined by tracing the spatial contour of the progress variable $\tilde{c}=0.5$ field. The progress variable $\tilde{c}$ represents the progress of the reaction from premixed reactants ($\tilde{c}=0$) to combustion products ($\tilde{c}=1$). This contour is identified by locating the maxima of Gaussian fits applied to radial slices across the image. Finally, the flame height is calculated as the distance between the maxima of the Gaussian fits along the contour and the nozzle exit, representing the flame's base. This methodology for computing flame height was discussed in our earlier study [27]. Figure 2 presents the influence of H_2 blending and varying turbulence intensity on the normalized flame height conditioned at a constant $T_{\text{ad}}=1800\pm20$ K. With increasing H_2 content as well as increasing turbulence intensity, the flame height decreases, leading to a longer postflame residence time for these cases at a fixed probe location.

Given this flame height, we then define the postflame residence time, τ_{res}, as

$$\tau_{\text{res}}=(x_{\text{probe}}-x_{\text{flame}})/U_{\text{prod}} \tag{1}$$

$$U_{\text{prod}}=U_0\frac{T_{\text{ad}}}{T_{\text{in}}} \tag{2}$$

where U_{prod} is the product gas velocity which is calculated using the bulk velocity U_0 and gas expansion ratio ($T_{\text{ad}}/T_{\text{in}}$). x_{probe} is the distance between nozzle exit and quartz probe tip. Note that this definition is well defined and reproducible, but not unique, and approaches could also be used. Also, note that τ_{res} can be negative if data points are taken below $\tilde{c}=0.5$.

3 Results and Discussion

3.1 Baseline CO and NO Plots. In Figs. 3 and 4, measured NO and CO levels (in ppmvd), respectively, from a CH_4–air flame at $\varphi=0.9$ ($T_{\text{ad}}=2130$ K) is presented as a function of τ_{res} (in milliseconds). The color bar in these plots indicates the turbulence intensity (u_{rms}/U_0) values. Figure 3 illustrates typical behavior of NO production with residence time—an initial jump (when extrapolating back to zero residence time) driven by flame NO formation, followed by a linear rise associated with thermal NO production. As also expected, CO levels decrease with increasing residence time due to the oxidation of CO to CO_2 in Fig. 4.

Next, constant T_{ad} slice is obtained across varying H_2 content. Here, both H_2 content and the equivalence ratio are varied simultaneously to achieve a constant T_{ad}. Figure 5 plots the CO emissions as a function of residence time across different H_2 blends at $T_{\text{ad}}=1800\pm20$ K, $U_0=10$ m/s, and turbulence intensity of 20%. In the case of CO, a significant reduction is observed as a function of $H_2\%$. For a fixed Tad, the variation in NO levels across H_2 blends falls within the uncertainty bounds making it insufficient to ascertain influence of H_2 blending and hence not shown here.

3.2 CO Emissions at Fixed Residence Time. In this section, CO emission levels are examined at a constant residence time (τ_{res}). This analysis underscores the impact of turbulence intensity on flame height and the influence of H_2 content on flame kinetics. These factors, in turn, affect the distribution of total τ_{res}s between the in-flame and postflame regions as discussed in Sec. 2.2. Figure 6 shows the CO emissions from a CH_4–air flame at $\varphi=0.9$ at fixed $\tau_{\text{res}}=0$–0.5 ms and 15 ms. At a fixed residence time, CO levels show a slight decrease as turbulence intensity increases. As the residence time τ_{res} has been corrected for the variation in flame height across turbulence intensities, the observed reduction in CO levels can be attributed to enhanced oxidation of CO to CO_2 facilitated by turbulent transport.

This analysis is now repeated for the $T_{\text{ad}}=1800\pm20$ K case in Fig. 7 presenting CO levels normalized by equilibrium CO for a given

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/4/041004/7534825/gtp-25-1459.pdf by guest on 31 December 2025

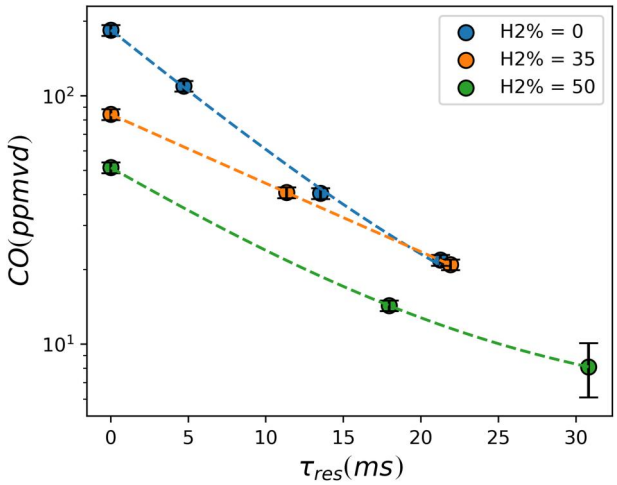

Fig. 5 CO levels (ppmvd) as a function of residence time for T_{ad}=1800±20 K at u_{rms}/U_0 = 20% and U_0=10 m/s

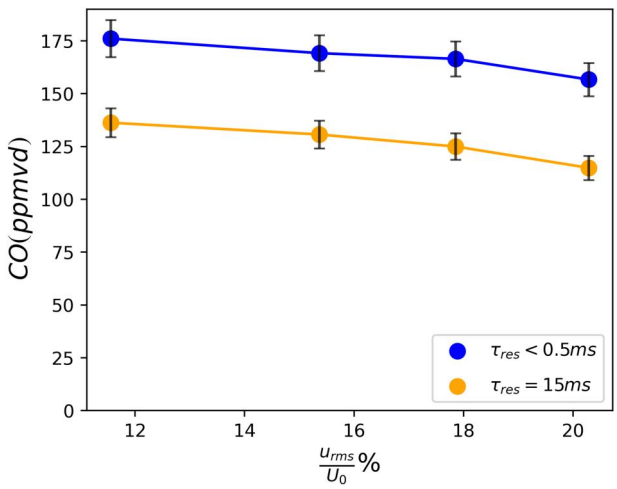

Fig. 6 CO emissions from CH_4 flame at φ=0.9 at fixed τ_{res} = 0–0.5 ms and 15 ms

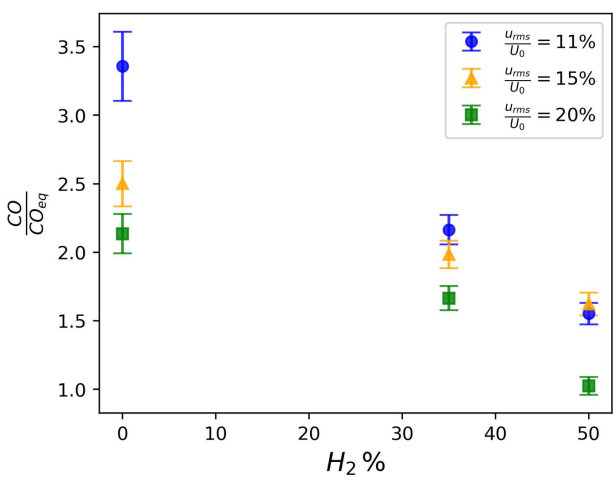

Fig. 7 CO normalized by equilibrium CO levels plotted as a function of H_2% in fuel at T_{ad}=1800±20 K at U_0=10 m/s, u_{rms}/U_0=11% (circles), u_{rms}/U_0=15% (triangles), and u_{rms}/U_0=20% (squares)

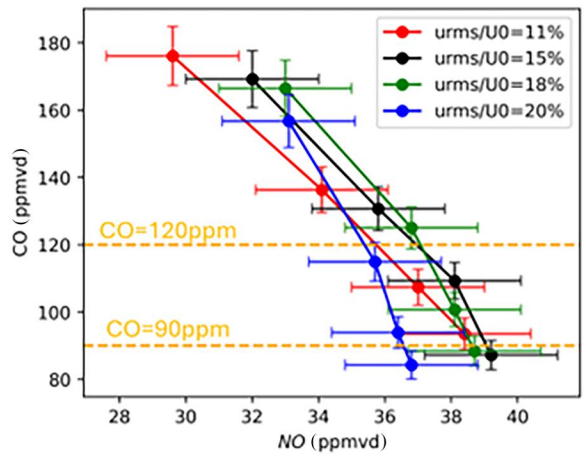

Fig. 8 NO levels plotted against CO levels for the four different turbulence intensity cases of 11%, 15%, 18%, and 20% for CH_4–air flame at φ=0.9 and U_0=10 m/s

case across turbulence intensities of 11% (circles), 15% (triangles), and 20% (squares). To facilitate readability, the constant residence time slices are staggered slightly as follows: 0 ms (u_{rms}/U_0 = 11%), 1.5 ms (u_{rms}/U_0 = 15%), and 3 ms (u_{rms}/U_0 = 20%). This visualization offers insights into the kinetic effects of H_2 blending, which in turn impacts CO reduction, distinguishing it from the net reduction in atomic carbon that typically contributes to lower CO levels. As this ratio varies as a function of H_2%, the observed variation in CO levels in Fig. 5 is not solely due to an absolute reduction in C atom concentration with increasing H_2 but also due to the accelerated conversion of CO to CO_2. For instance, the CO emissions are about 3.4 times their equilibrium value at 0% H_2 case as compared to being at 1.5 times equilibrium value at the 50% H_2 case. With sufficiently long residence time, the measured CO value attains its equilibrium levels across conditions. Note that with increasing turbulence intensity, reduction in CO levels is observed with corrected or postflame residence time values.

3.3 NO Emissions at Fixed CO. While attempting to interpolate results to fixed residence time is helpful, it still requires introducing a flame height definition. An alternative approach is to plot NO results at fixed CO levels. As such, relationships between the NO and CO levels are visualized in this section. Of particular

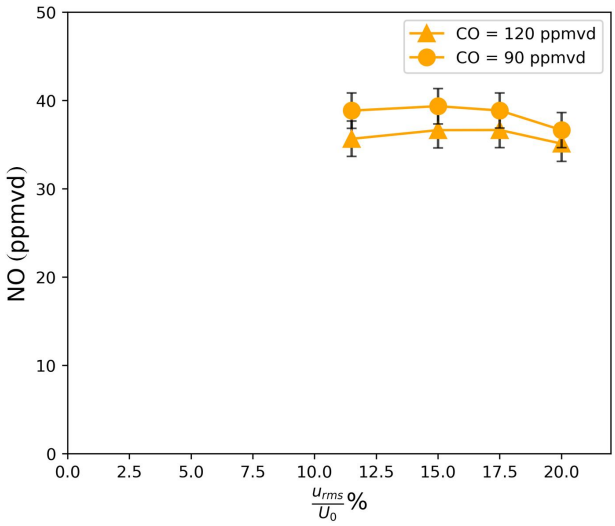

Fig. 9 NO levels plotted as a function of turbulence intensity conditioned at constant CO values of 90 ppmvd (circles) and 120 ppmvd (triangles)

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/4/041004/7534825/gtp-25-1459.pdf by guest on 31 December 2025

interest here is the effect of turbulence intensity on NO emissions. Figure 8 represents NO and CO levels from Figs. 3 and 4 plotted against each other with data grouped based on the turbulence intensity %: 11%, 15%, 18%, and 20% .

To gain insights into the effect of turbulence intensity, NO levels at constant CO values of 90 and 120 ppmvd are presented in Fig. 9. These data slices are inspired by the typical NO_x–CO tradeoff curves where optimal τ_{res} is chosen based on both NO_x and CO levels simultaneously as they, respectively, increase and decrease with τ_{res}. From Fig. 9, as a function of turbulence intensity, the effect on NO levels remains within the uncertainty bounds, indicating that NO does not vary significantly with turbulence intensity.

4 Conclusions

To advance the development of next-generation gas turbine combustors designed for high-hydrogen fuel blends, we conducted a fundamental study exploring the effects of incoming flow turbulence and fuel hydrogen content on NO_x and CO emissions. H_2 exhibits markedly distinct combustion characteristics compared to hydrocarbons, including significantly higher flame temperatures at equivalent equivalence ratios. To enable systematic comparisons, measurements were performed at constant adiabatic flame temperatures across varying conditions. Additionally, the postflame residence time was corrected to account for the influence of hydrogen content and turbulence intensity on flame height, as increases in either parameter result in reduced flame height. Data from this study reveal that NO levels (in ppmvd) at fixed CO levels do not exhibit significant variation with increasing turbulence intensity. Therefore, while local NO_x production may increase with higher H_2 concentrations—potentially due to superadiabatic zones forming at flame front cusps—this effect is expected to be offset by regions of lower NO_x production. As a result, overall NO_x emissions remain unchanged with increasing turbulence intensity. Additionally, both increased turbulence intensity and higher H_2 content in the fuel lead to a reduction in CO levels at a constant residence time. The decrease in CO with increased turbulence intensity is attributed to enhanced oxidation of CO to CO_2, while the reduction in CO with higher H_2 content is due to both a kinetic effect as well as a decrease in the carbon atoms.

Funding Data

- Department of Energy (Contract No. DE-FE0032079).
- EPRI (Contract No. DKT230956).

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

BR = blockage ratio
CO_{eq} = equilibrium CO levels
D = burner nozzle diameter
T_{ad} = adiabatic flame temperature
T_{in} = preheat temperature
u_{rms} = RMS of velocity fluctuation
U_0 = bulk flow velocity
τ_{res} = postflame residence time
φ = equivalence ratio

References

[1] International Energy Agency, 2023, "Global Hydrogen Review 2023," IEA, Paris, France, accessed Sept. 19, 2025, https://www.iea.org/reports/global-hydrogen-review-2023

[2] Kalantari, A., Sullivan-Lewis, E., and McDonell, V., 2016, "Flashback Propensity of Turbulent Hydrogen–Air Jet Flames at Gas Turbine Premixer Conditions," ASME J. Eng. Gas Turbines Power, **138**(6), p. 061506.

[3] Lieuwen, T., McDonell, V., Petersen, E., and Santavicca, D., 2008, "Fuel Flexibility Influences on Premixed Combustor Blowout, Flashback, Autoignition, and Stability," ASME. J. Eng. Gas Turbines Power., **130**(1), p. 011506.

[4] Karim, G., Wierzba, I., and Al-Alousi, Y., 1996, "Methane-Hydrogen Mixtures as Fuels," Int. J. Hydrogen Energy, **21**(7), pp. 625–631.

[5] Milton, B., and Keck, J., 1984, "Laminar Burning Velocities in Stoichiometric Hydrogen and Hydrogenî—, Hydrocarbon Gas Mixtures," Combust. Flame, **58**(1), pp. 13–22.

[6] Jackson, G. S., Sai, R., Plaia, J. M., Boggs, C. M., and Kiger, K. T., 2003, "Influence of H2 on the Response of Lean Premixed CH4 Flames to High Strained Flows," Combust. Flame, **132**(3), pp. 503–511.

[7] Day, M., Bell, J., Bremer, P.-T., Pascucci, V., Beckner, V., and Lijewski, M., 2009, "Turbulence Effects on Cellular Burning Structures in Lean Premixed Hydrogen Flames," Combust. Flame, **156**(5), pp. 1035–1045.

[8] Venkateswaran, P., Marshall, A., Seitzman, J., and Lieuwen, T., 2015, "Scaling Turbulent Flame Speeds of Negative Markstein Length Fuel Blends Using Leading Points Concepts," Combust. Flame, **162**(2), pp. 375–387.

[9] Singh, S., Rajagopalan, H., Emerson, B., Noble, D., Steele, R., and Lieuwen, T. C., 2025, "Numerical Investigation of CO and NO Production From Premixed Hydrogen/Methane Fuel Blends," ASME J. Eng. Gas Turbines Power, **147**(4), p. 041015.

[10] Shy, S., Chen, Y., Yang, C., Liu, C., and Huang, C., 2008, "Effects of H2 or CO2 Addition, Equivalence Ratio, and Turbulent Straining on Turbulent Burning Velocities for Lean Premixed Methane Combustion," Combust. Flame, **153**(4), pp. 510–524.

[11] Venkateswaran, P., Marshall, A., Seitzman, J., and Lieuwen, T., 2013, "Pressure and Fuel Effects on Turbulent Consumption Speeds of H2/CO Blends," Proc. Combust. Inst., **34**(1), pp. 1527–1535.

[12] Marshall, A., Lundrigan, J., Venkateswaran, P., Seitzman, J., and Lieuwen, T., 2017, "Measurements of Stretch Statistics at Flame Leading Points for High Hydrogen Content Fuels," ASME J. Eng. Gas Turbines Power, **139**(11), p. 111503.

[13] Godbold, C., Breer, B., Rajagopalan, H. P., Miller, J., Bond, R., Emerson, B., Wu, D., Acharya, V., Noble, D., and Lieuwen, T., 2023, "Development of a NOx Formation Test Facility for Lean Premixed Methane-Hydrogen-Air Flames,".

[14] Skottene, M., and Rian, K. E., 2007, "A Study of NOx Formation in Hydrogen Flames," Int. J. Hydrogen Energy, **32**(15), pp. 3572–3585.

[15] Lin, Y.-C., Daniele, S., Jansohn, P., and Boulouchos, K., 2012, "Combustion Characteristics and NOx Emission of Hydrogen-Rich Fuel Gases at Gas Turbine Relevant Conditions," ASME Paper No. GT2012-69080.

[16] Douglas, C. M., Shaw, S. L., Martz, T. D., Steele, R. C., Noble, D. R., Emerson, B. L., and Lieuwen, T. C., 2022, "Pollutant Emissions Reporting and Performance Considerations for Hydrogen–Hydrocarbon Fuels in Gas Turbines," ASME J. Eng. Gas Turbines Power, **144**(9), p. 091003.

[17] Therkelsen, P., Werts, T., McDonell, V., and Samuelsen, S., 2009, "Analysis of NOx Formation in a Hydrogen-Fueled Gas Turbine Engine," ASME. J. Eng. Gas Turbines Power., **131**(**3**), p. 031507.

[18] Cellek, M. S., and Pınarbaşı, A., 2018, "Investigations on Performance and Emission Characteristics of an Industrial Low Swirl Burner While Burning Natural Gas, Methane, Hydrogen-Enriched Natural Gas and Hydrogen as Fuels," Int. J. Hydrogen Energy, **43**(2), pp. 1194–1207.

[19] Breer, B., Rajagopalan, H., Godbold, C., Johnson, H., II, Emerson, B., Acharya, V., Sun, W., Noble, D., and Lieuwen, T., 2023, "Numerical Investigation of NOx Production From Premixed Hydrogen/Methane Fuel Blends," Combust. Flame, **255**, p. 112920.

[20] Correa, S. M., 1993, "Turbulence-Chemistry Interactions in the Intermediate Regime of Premixed Combustion," Combust. Flame, **93**(1–2), pp. 41–60.

[21] Cheng, R., Littlejohn, D., Strakey, P., and Sidwell, T., 2009, "Laboratory Investigations of a Low-Swirl Injector With H2 and CH4 at Gas Turbine Conditions," Proc. Combust. Inst., **32**(2), pp. 3001–3009.

[22] Cheng, L., Chen, Y., Pei, Y., Sun, G., Zou, J., Peng, S., and Zhang, Y., 2024, "NO and CO Emission Characteristics of Laminar and Turbulent Counterflow Premixed Hydrogen-Rich Syngas/Air Flames," Processes, **12**(3), p. 475.

[23] Day, M. S., Bell, J. B., Gao, X., and Glarborg, P., 2011, "Numerical Simulation of Nitrogen Oxide Formation in Lean Premixed Turbulent H2/O2/N2 Flames," Proc. Combust. Inst., **33**(1), pp. 1591–1599.

[24] Marshall, A., Venkateswaran, P., Noble, D., Seitzman, J., and Lieuwen, T., 2011, "Development and Characterization of a Variable Turbulence Generation System," Exp. Fluids, **51**(3), pp. 611–620.

[25] Colket, M., III, Chiappetta, L., Guile, R., Zabielski, M., and Seery, D., 1982, "Internal Aerodynamics of Gas Sampling Probes," Combust. Flame, **44**(1–3), pp. 3–14.

[26] Goodwin, D. G., Speth, R. L., Moffat, H. K., and Weber, B. W., 2017, "Cantera: An Object-Oriented Software Toolkit for Chemical Kinetics, Thermodynamics, and Transport Processes," Zenodo, Geneva, Switzerland, accessed Sept. 19, 2025, https://doi.org/10.5281/zenodo.170284

[27] Rajagopalan, H. P., Cole, R., Wu, D., Emerson, B., and Lieuwen, T. C., 2025, "Turbulent Burning Velocity of High Hydrogen Flames," ASME J. Eng. Gas Turbines Power, **147**(1), p. 011016.

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/4/041004/7534825/gtp-25-1459.pdf by guest on 31 December 2025